

ROHIT PANCHAL

Software Engineer | Backend, Cloud and Applied AI

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SUMMARY

Software engineer with 4+ years at Krishify, building backend systems, cloud infrastructure, and applied AI for an agritech platform serving **9M+ farmers**. Lead technical design and mentor a **4-engineer backend team**. Built the company's GCP infrastructure, reduced monthly cloud spend by **62.5%**, and shipped a tenant-aware LLM analytics agent.

EXPERIENCE

Krishify

Software Engineer

Software Engineer Intern

Gurugram, India

Aug 2022 – Present

Mar 2022 – Jul 2022

Backend Architecture and Scale

- Own APIs, database schemas, and microservices for the farmer app (**1.5M monthly active users**), operator app, and B2B SaaS platform handling **~10M requests/day**. Enterprise clients include AVT McCormick, AB InBev, Swiggy, and Rallis India (a Tata company).
- Designed and shipped the SaaS architecture for **10+ enterprise tenants**, defining service boundaries, tenant-aware database routing, and shared versus dedicated deployments; reduced tenant onboarding from ~4 weeks to **under 5 days**.
- Re-architected services as stateless, autoscaled GKE workloads with health probes and graceful shutdown, supporting 5× seasonal traffic spikes, zero-downtime rolling deployments, and **99.9% availability** over the past 12 months.
- Halved p95 API latency from ~1.6 s to **~800 ms** and reduced Cloud SQL connections by **~60%** during 3× workload growth through query-plan analysis, indexing, connection pooling, and Redis caching (~85% hit rate on hot feed reads).
- Migrated Python 3.7 to 3.11, Django 2.2 to 5.2, and DRF 3.9 to 3.16 with zero production incidents; reduced mean request time by ~20% and cleared ~30 end-of-life or CVE-flagged dependencies.

Applied AI

- Designed and shipped **Ceres**, a natural-language analytics agent used by 2 enterprise clients and ~120 weekly active users. Built the streaming FastAPI/Pydantic AI backend, React chat UI, provider-agnostic LLM layer, and **150-case regression evaluation** with ~92% query accuracy.
- Built an **internal proof of concept** for multilingual AI calling bots using Pipecat, Sarvam AI speech-to-text/text-to-speech, Claude, and Tata SmartFlo for farmer conversations.
- Authored the build plan for an agriculture-domain small language model using proprietary data from **9M+ farmers**, evaluating Indic foundation models and fine-tuning economics.

Cloud Infrastructure and Delivery

- Architected and built GCP infrastructure from scratch with a multi-project, multi-region Terraform monorepo provisioning GKE and Cloud SQL for PostgreSQL.
- Reduced monthly cloud spend by **62.5%**, from INR 12 lakh to INR 4.5 lakh (**INR 90 lakh in annualized savings**), without capacity or SLO regression. Consolidated workloads onto GCP, retired 7 redundant services, rightsized GKE and Cloud SQL, and rebuilt BigQuery-to-RDS pipelines.
- Built self-service Bitbucket Pipelines and ArgoCD GitOps delivery for backend, frontend, and Terraform across GKE and EKS, enabling independent releases with deployment lead time under 10 minutes.
- Drove infrastructure readiness for ISO 27001 certification through cross-region backups, disaster recovery, automated credential rotation, SSO-based access management, and encryption at rest.

Security Engineering

- Deployed CrowdStrike Falcon across 2 GKE clusters (~12 nodes) and bastion VMs, implementing node sensors, Kubernetes admission control, and registry image assessment; blocked vulnerable images at deployment and achieved 100% runtime coverage of production workloads.
- Hardened Android-to-backend APIs with TLS enforcement, certificate pinning compatible with cert-manager rotation, Google Play Integrity, HMAC request signing, and Cloud Armor WAF policies.

Data and Product Engineering

- Led Google Earth Engine initiatives using NDVI, NDWI, and LAI for harvest-date and crop-disease prediction, achieving ± 5 -day harvest-date accuracy across 3 crops; built Farmwise APIs delivering crop insights to **3,000+ field operators**.
- Increased feed click-through rate from ~1.3% to 4% (~3×) in a 90-day A/B test in 2023. Built 4 segmentation personas that improved campaign conversion by 75% over untargeted sends.

TECHNICAL SKILLS

Languages and frameworks: Python, SQL, Bash, Django, Django REST Framework, FastAPI, React (AI-assisted)

Architecture and data: System design, multi-tenant SaaS, REST APIs, microservices, PostgreSQL, Redis, Celery, Elasticsearch, BigQuery, Google Earth Engine

Cloud and DevOps: GCP (GKE, Cloud SQL, IAM, Cloud Armor), AWS (EKS, RDS), Terraform, Kubernetes, Docker, Bitbucket Pipelines, ArgoCD, ingress-nginx, cert-manager, Prometheus, Grafana, Cloud Monitoring

AI and LLMs: Claude API, Pydantic AI, agent evaluation, RAG, Pipecat, Sarvam AI, Tata SmartFlo, Claude Code, multi-model agent workflows

Security: CrowdStrike Falcon, TLS/PKI, JWT, SSO, HMAC, Google Play Integrity, least-privilege IAM, certificate pinning

EDUCATION

B.Tech in Computer Science and Engineering

2018 – 2022

Guru Gobind Singh Indraprastha University, New Delhi | CGPA: 8.81/10